

- ① In the population the average IQ is 100. A team of Researchers want to test a new medication to see if it has either a positive or negative impact on Intelligence or no impact at all. A sample of 30 participants who have taken the medication has a mean of 140 with a standard deviation of 20.

Did the medication effect Intelligence ($\bar{x} = 95\%$, $\alpha = 0.05$)

Ans

$$\mu = 100$$

$$n = 30$$

$$\bar{x} = 140$$

$$s = 20$$

z-test ($\bar{x}$)

↓
t-test (s)

① Null hypothesis (H_0) = $\mu = 100$

② Alternate hypothesis (H_1) = $\mu \neq 100$ 2 Tail test

$$\alpha = 0.05$$

③ Degree of freedom

$$df = n - 1 \Rightarrow 30 - 1 \Rightarrow 29$$

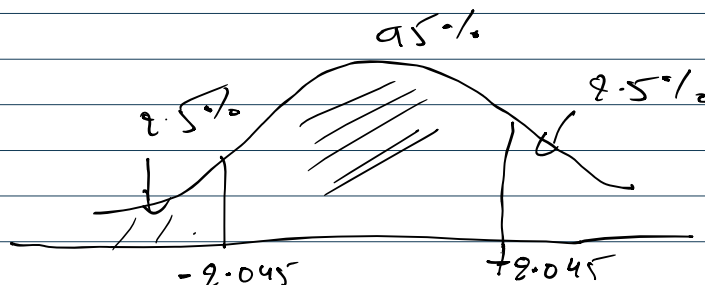
$$df: 29$$

$$df = 29$$

$$\alpha = 0.05$$

2 tail test

④



If t-test is less than -2.045 or >2.045 we will reject the null hypothesis

⑤ Calculate test Statistics

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{140 - 100}{20/\sqrt{30}}$$

$$= \frac{40}{3.65} = 10.96$$

$$t = 10.96$$

Conclusion/Inference

$$\mu \neq 100$$

Since t -Value $10.96 > 2.045$ so the medication is showing an effect on intelligence (H_1 is true)